

21MDS81

DATA PRIVACY AND SECURITY ANALYTICS

(Unit IV)

Outline

- Privacy considerations in machine learning
- Privacy complications in the AI era
- Threats and attacks for ML systems
- Privacy-preserving machine learning

Privacy considerations in ML

- Our search queries, browsing history, purchase transactions, watched videos, and movie preferences are a few types of information that are collected and stored daily.
- Advances in artificial intelligence have increased the ability to capitalize on and benefit from the collection of private data.
- This data collection happens within our mobile devices and computers, on the streets, and even in our own offices and homes, and the data is used by a variety of machine learning (ML) applications in different domains, such as marketing, insurance, financial services, mobility, social networks, and healthcare.

Privacy considerations in ML (Contd...)

- More and more cloud-based data-driven ML applications are being developed by different service providers (who can be classified as the *data users*, such as Facebook, LinkedIn, and Google).
- Most of these applications leverage the vast amount of data collected from each individual (the *data owner*) to offer users valuable services.
- These services often give users some commercial or political advantage by facilitating various user recommendations, activity recognition, health monitoring, targeted advertising, or even election predictions.
- However, on the flip side, the same data could be repurposed to infer sensitive (private) information, which would jeopardize the privacy of individuals.
- Moreover, with the increased popularity of Machine Learning as a Service (MLaaS), where cloud-based ML and computing resources are bundled together to provide efficient analytical platforms (such as Microsoft Azure Machine Learning Studio, AWS Machine Learning, and Google Cloud Machine Learning Engine), it is necessary to take measures to enforce privacy on those services before they are used with sensitive datasets.

Privacy complications in the AI era

Facebook–Cambridge Analytica scandal in April 2018

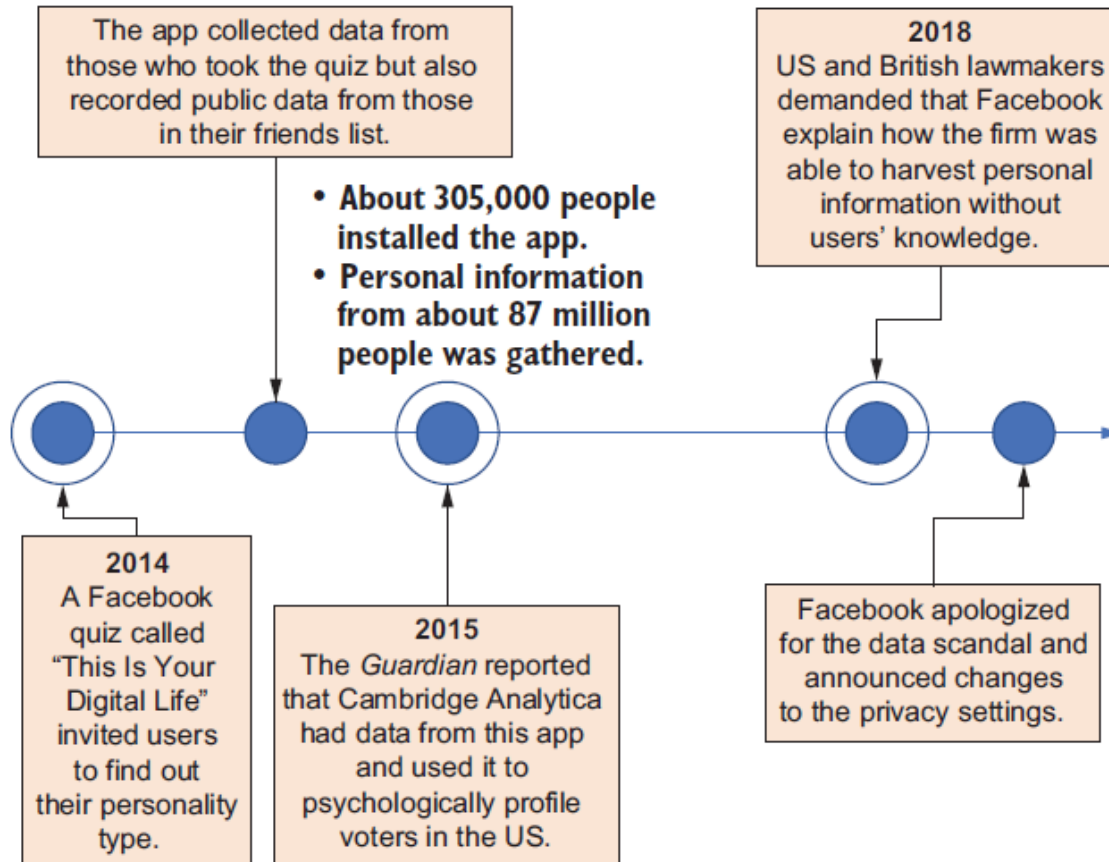


Figure 1.1 The Facebook–Cambridge Analytica scandal raised the alarm about privacy concerns.

Threat of learning beyond the intended purpose

- ML can be seen as the capability of an algorithm to mimic intelligent human behavior, performing complex tasks by looking at data from different angles and analyzing it across domains.
- This learning process is utilized by various applications in our day-to-day life, from product recommendation systems in online web portals to sophisticated intrusion detection mechanisms in internet security applications.

Threat of learning beyond the intended purpose (Contd...)

Use of private data on the fly

- ML applications require vast amounts of data from various sources to produce high confidence results.
- Web search queries, browsing histories, transaction histories of online purchases, movie preferences, and individual location check-ins are some of the information collected and stored on a daily basis, usually without users' knowledge.
- Some of this **information** is **private** to the individuals, but it is **uploaded to high-end centralized servers** mostly in **clear-text format** so **ML algorithms can extract patterns** and build ML models from it.

Threat of learning beyond the intended purpose (Contd...)

- The problem is not limited to the collection of this private data by different ML applications.
- The data is also exposed to insider attacks because the information is available to workers at data-mining companies.
- For example, database administrators or application developers may have access to this data without many restrictions.
- This data may also be exposed to external hacking attacks, with private information being revealed to the outside world.
- For instance, in 2015 Twitter fired an engineer after intelligence officials found that he might have been spying on the accounts of Saudi dissidents by accessing user details such as phone numbers and IP addresses.

Threat of learning beyond the intended purpose (Contd...)

How data is processed inside ML algorithms

- To understand the relationship between data privacy and ML algorithms, knowing how ML systems work with the data they process is crucial.
- Typically, we can represent the input data for an ML algorithm (captured from various sources) as a set of sample values, and each sample can be a group of features.
- Let's take the example of a facial recognition algorithm that recognizes people when they upload an image to Facebook.
- Consider an image of 100×100 pixels where a single value from 0 to 255 represents each pixel. These pixels can be concatenated to form a feature vector.
- Each image can be represented to the ML algorithm as a vector of data, along with an associated label for that data.
- The ML algorithm would use multiple feature vectors and their associated labels during the training phase to produce an ML model.
- This model would then be used with fresh, unseen data (testing samples) to predict the result—in this case, to recognize a person.

Threat of learning beyond the intended purpose (Contd...)

Why privacy protection in ML is important

- When personal information is used for a wrong or unintended purpose, it can be manipulated to gain a competitive advantage.
- When massive volumes of personal records are coupled with ML algorithms, no one can predict what new results they may produce or how much private information those results may reveal.
- The Facebook–Cambridge Analytica scandal discussed earlier is a perfect example of the wrongful use of personal data.

Threat of learning beyond the intended purpose (Contd...)

- Hence, when designing ML algorithms for an application, ensuring privacy protection is vital.
- First, it ensures that other parties (data users) cannot use the personal data for their own advantage.
- Second, everyone has things that they do not want others to know.
- For example, they may not want others to know the details of their medical history. But ML applications are data driven, and we need training samples to build a model. We want to use the private data to build a model, but we want to prevent the ML algorithm from learning anything sensitive. How can we do that?

Threat of learning beyond the intended purpose (Contd...)

- Let's consider a scenario where we use two databases:
 - a sanitized medical database that lists patients' histories of medication prescriptions,
 - another data source with user information and pharmacies visited.
- When these sources are linked together, the correlated database can have additional knowledge, such as which patient bought their medication from which pharmacy.
- Suppose we are using this correlated dataset with an ML application to extract relationships between the patients, medications, and pharmacies.
- While it will extract the obvious relations between different diseases and the medications prescribed, it may also learn roughly where the patient resides simply by referring to the zip codes of their most-visited pharmacies, even if the data does not contain patient addresses.
- This is a simple example, but you can imagine how severe the consequences could be if privacy is not protected.

Threat of learning beyond the intended purpose (Contd...)

Regulatory requirements and the utility vs. privacy tradeoff

- Traditionally, data security and privacy requirements were set by the data owners (such as organizations) to safeguard the competitive advantage of the products and services they offered.
- However, in the big data era, data has become the most valuable asset in the digital economy, and governments imposed many privacy regulations to prevent the use of sensitive information beyond its intended purpose.
- Privacy standards such as HIPAA (Health Insurance Portability and Accountability Act of 1996), PCI DSS (Payment Card Industry Data Security Standard), FERPA (Family Educational Rights and Privacy Act), and the European Union's GDPR (General Data Protection Regulation) are some of the privacy regulations that organizations commonly adhere to.

Threat of learning beyond the intended purpose (Contd...)

- For example, regardless of the size of practice, most healthcare providers transmit health information electronically, such as for claims, medication records, benefit eligibility inquiries, referral authorization requests, and the like.
- However, HIPAA regulations require that these healthcare providers protect sensitive patient health information from being disclosed without the patient's consent or knowledge.

Threat of learning beyond the intended purpose (Contd...)

- Regardless of whether the data is labeled or not, or whether raw data is preprocessed, ML models are essentially very sophisticated statistics based on the training dataset, and ML algorithms are optimized to squeeze every bit of utility out of the data.
- Therefore, in most conditions, they are capable of learning sensitive attributes in the dataset, even when that is not the intended task.
- When we attempt to preserve privacy, we want to prevent these algorithms from learning sensitive attributes.
- Hence, as you can see, utility and privacy are on opposite ends of the spectrum. When you tighten privacy, it can affect the performance of the utility.

Threat of learning beyond the intended purpose (Contd...)

- The real challenge is **balancing privacy and performance** in ML applications so that we can better utilize the data while ensuring the privacy of the individuals.
- Because of the regulatory and application-specific requirements, we cannot degrade privacy protection just to increase the utility of the application.
- On the other hand, privacy has to be implemented systematically without using arbitrary mechanisms, as many additional threats must be considered.
- ML applications are prone to different privacy and security attacks.
- We will explore these potential attacks in detail next and look at how we can mitigate them by designing privacy-preserving ML (PPML) algorithms.

Threats and Attacks for ML Systems

- Figure 1.2 is a timeline showing a list of threats and attacks for ML systems, including de-anonymization (re-identification) attacks, reconstruction attacks, parameter inference attacks, model inversion attacks, and membership inference attacks.

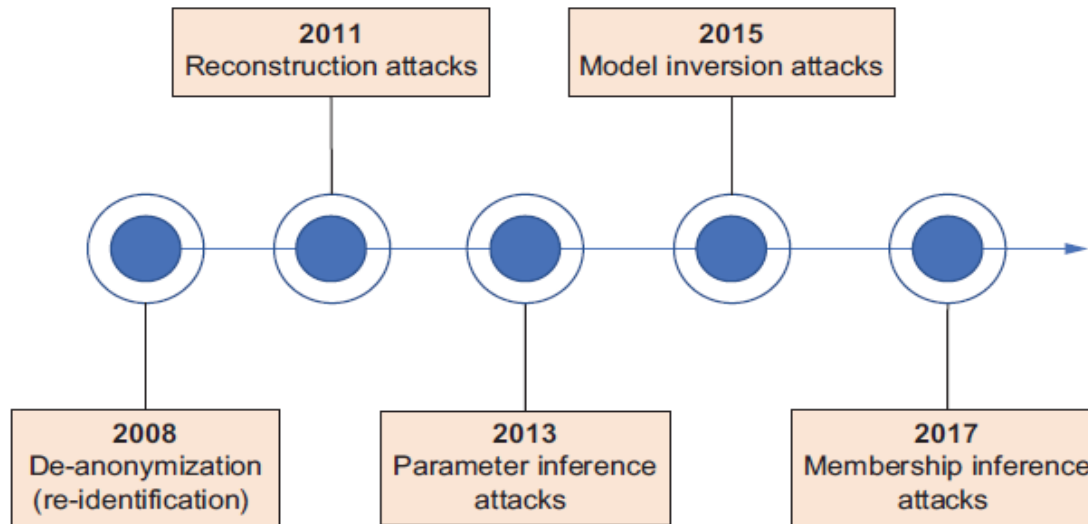


Figure 1.2 A timeline of threats and attacks identified for ML systems

- Although some leading companies, such as Google and Apple, started designing and utilizing their own privacy-preserving methodologies for ML tasks, it is still a challenge to improve public awareness of these privacy technologies.

Threats and Attacks for ML Systems (Contd...)

The problem of private data in the clear

- Figure 1.3 illustrates a typical client/server application scenario.
- When an application collects private information, that information is often transferred, possibly through encrypted channels, to cloud-based servers where the learning happens.
- In the figure, a mobile application connects to a cloud server to perform an inference task.

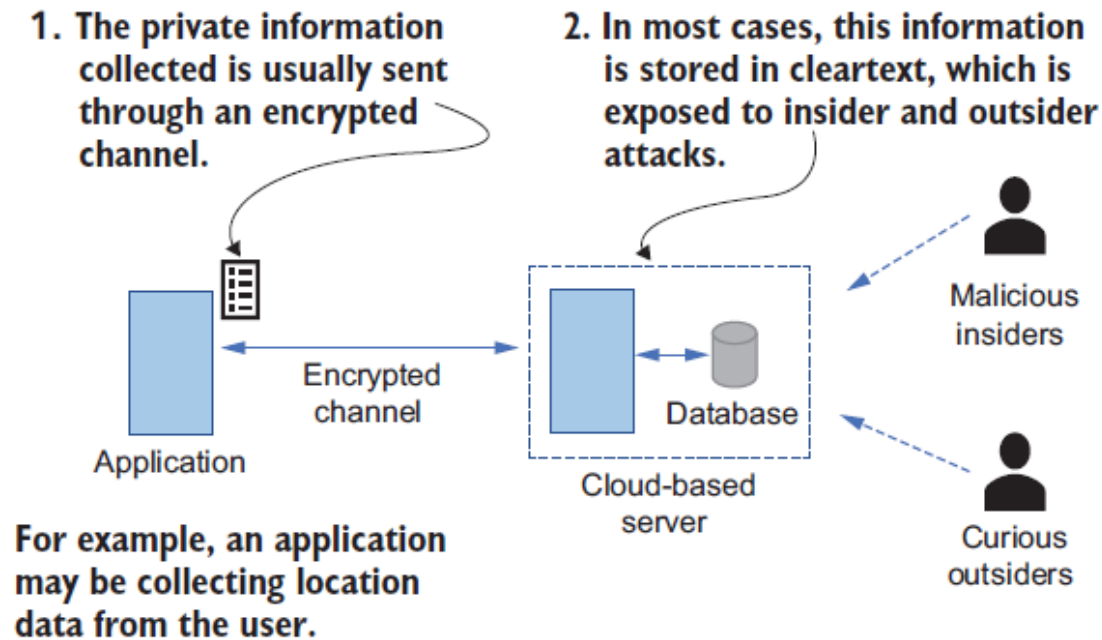


Figure 1.3 The problem of storing private data in cleartext format

Threats and Attacks for ML Systems (Contd...)

- For example, a parking app may send the user's location data to find a nearby available garage.
- Even though the communication channel is secured, the data most likely resides in the cloud in its original unencrypted form or as features extracted from the original record.
- This is one of the biggest challenges to privacy because that data is susceptible to various insider and outsider attacks.

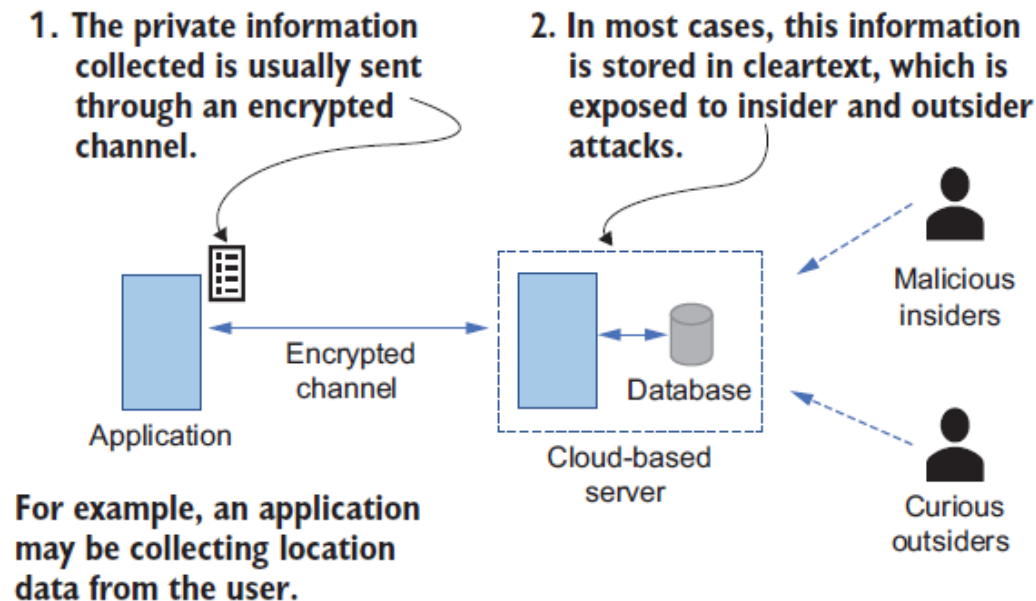


Figure 1.3 The problem of storing private data in cleartext format

Threats and Attacks for ML Systems (Contd...)

Reconstruction attacks

- It is essential to store private data on the server in encrypted form, and we should not send raw data directly to the server in its original form.
- However, reconstruction attacks pose another possible threat: the attacker could reconstruct data even without having access to the complete set of raw data on the server.
- In this case, the adversary gains an advantage by having external knowledge of feature vectors (the data used to build the ML model).
- The adversary usually requires direct access to the ML models deployed on the server, which is referred to as white-box access (see table 1.1).
- They then try to reconstruct the raw private data by using their knowledge of the feature vectors in the model. These attacks are possible when the feature vectors used during the training phase to build the ML model are not flushed from the server after building the model.

Threats and Attacks for ML Systems (Contd...)

Table 1.1 Difference between white-box, black-box, and grey-box access

White-box access	Black-box access	Grey-box access
Has full access to the internal details of the ML models, such as parameters and loss functions	Has no access to the internal details of the ML models	Has partial access to the internal details of the ML models

Threats and Attacks for ML Systems (Contd...)

Example for Reconstruction Attacks:

- Reconstruction of fingerprint images from a minutiae template
- Reconstruction attacks against mobile-based continuous authentication systems

Threats and Attacks for ML Systems (Contd...)

RECONSTRUCTION OF FINGERPRINT IMAGES FROM A MINUTIAE TEMPLATE

- Fingerprint based authentication is prevalent in many organizations.
- Users are authenticated by comparing a newly acquired fingerprint image with a fingerprint image already saved in the user authentication system.
- In general, these fingerprint-matching systems use one of four different representation schemes known as
 - *Grayscale*
 - *Skeleton*
 - *phase* and
 - *Minutiae*

Threats and Attacks for ML Systems (Contd...)

Minutiae-based representation

- It is the most widely adopted one due to the compactness of the representation.
- Because of this compactness, many people erroneously think that the minutiae template does not contain enough information for attackers to reconstruct the original fingerprint image.
- In 2011, a team of researchers successfully demonstrated an attack that could reconstruct fingerprint images directly from minutiae templates.
- They reconstructed a phase image from minutiae, which was then converted into the original (grayscale) image.
- Next, they launched an attack against fingerprint recognition systems to infer private data.

Threats and Attacks for ML Systems (Contd...)

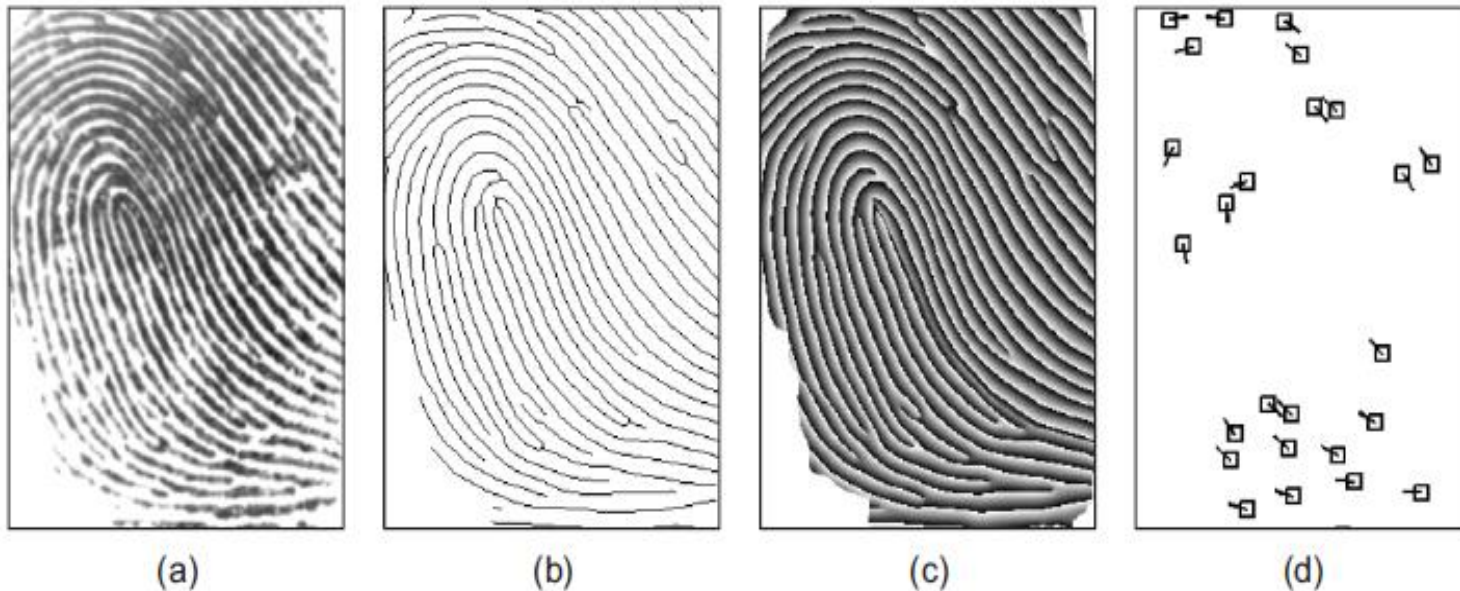


Figure 1.4 The four different types of representation schemes used in fingerprint matching systems: (a) grayscale image, (b) skeleton image, (c) phase image, and (d) minutiae

Threats and Attacks for ML Systems (Contd...)

RECONSTRUCTION ATTACKS AGAINST MOBILE-BASED CONTINUOUS AUTHENTICATION SYSTEMS

- Al-Rubaie et al. investigated the possibility of reconstruction attacks that used gestural raw data from users' authentication profiles in mobile-based continuous authentication systems.
- Continuous authentication is a method of verifying the user not just once but continuously throughout an entire session (such as Face ID in iPhones).
- Without continuous authentication, organizations are more vulnerable to many attack vectors, such as a system being taken over when it is no longer being used but the session remains open.
- In such a scenario, a reconstruction attack could use available private information that is leaked to the adversary.
- At a high level, Al-Rubaie et al. used the feature vectors stored in user profiles to reconstruct the raw data and then used that information to hack into other

Threats and Attacks for ML Systems (Contd...)

Model inversion attacks

- While some ML models store explicit feature vectors, other ML algorithms (such as neural networks and ridge regression) do not keep feature vectors inside the model.
- In these circumstances, the adversary's knowledge is limited, but they may still have access to the ML model (white-box access scenario).
- In another black-box access scenario, the adversary does not have direct access to the ML model: they can listen for incoming requests to an ML model when a user submits new testing samples and for responses generated by the model.
- In model inversion attacks, the adversary utilizes the responses generated by the ML model in a way that resembles the original feature vectors used to create the ML model.

Threats and Attacks for ML Systems (Contd...)

Figure 1.5 illustrates how a model inversion attack works.

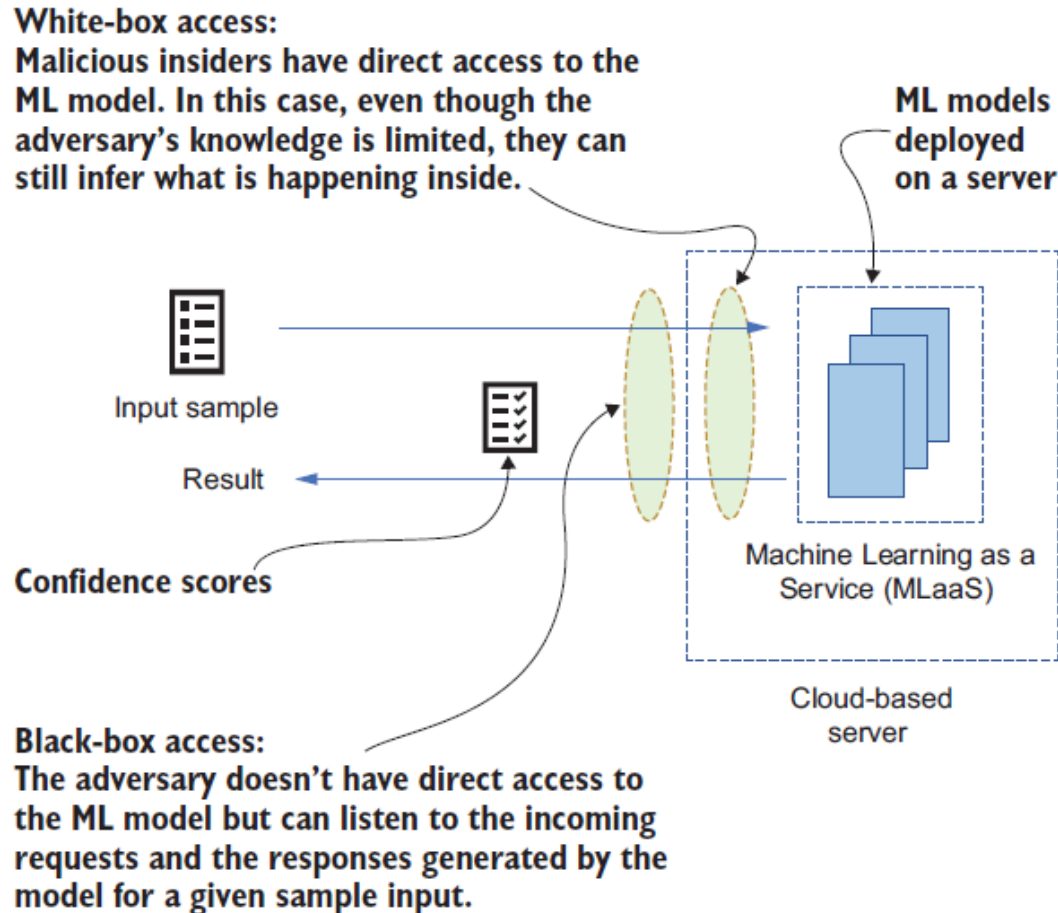


Figure 1.5 The difference between white-box access and black-box access. White-box access requires direct access and permission for the ML model to infer data; black-box access usually involves listening to the communication channel.

Threats and Attacks for ML Systems (Contd...)

- Model inversion attacks utilize the confidence values received from the ML model (such as the probability decision score) to generate feature vectors.
- For example, let's consider a facial recognition algorithm. When you submit a face image to the algorithm, the algorithm produces a result vector with the class and the corresponding confidence score based on the features it identifies in that image. For now, assume the result vector generated by the algorithm is

[John:.99, Simon:.73, Willey:.65]

- The meaning of this result is that, the algorithm is 99% confident (the decision score) that this image belongs to John (the class) and 73% confident that it belongs to Simon, and so on.

Threats and Attacks for ML Systems (Contd...)

What if the adversary can listen to all these communications?

- Even though they do not have the input image or know whose image it is, they can deduce that they will get a confidence score in this range if they input a similar image.
- By accumulating results over a certain period, the attacker can produce an average score representing a certain class in the ML model.
- If the class represents a single individual, as in a facial recognition algorithm, identifying the class could result in a threatening privacy breach.
- At the beginning of the attack, the adversary does not know who the person is, but by accumulating data over time, they will be able to identify the person, which is serious.
- Therefore, the model inversion attack is a severe threat to ML-based systems.

Threats and Attacks for ML Systems (Contd...)

- In mitigating model inversion attacks, limiting the adversary to black-box access is important, because it limits the adversary's knowledge.
- In our example of face recognition-based authentication, instead of providing the exact confidence value of a certain ML class, we could round it.
- Alternatively, only the final predicted class label may be returned, so that it is harder for an adversary to learn anything beyond the specific prediction.

Threats and Attacks for ML Systems (Contd...)

Membership inference attacks

- While model inversion attacks do not try to reproduce an actual sample from the training dataset, membership inference attacks try to infer a sample based on the ML model output to identify whether it was in the original training dataset.
- The idea behind a membership inference attack is that given an ML model, a sample, and domain knowledge, the adversary can determine whether the sample is a member of the training dataset used to build the ML model, as shown in figure 1.6.

Threats and Attacks for ML Systems (Contd...)

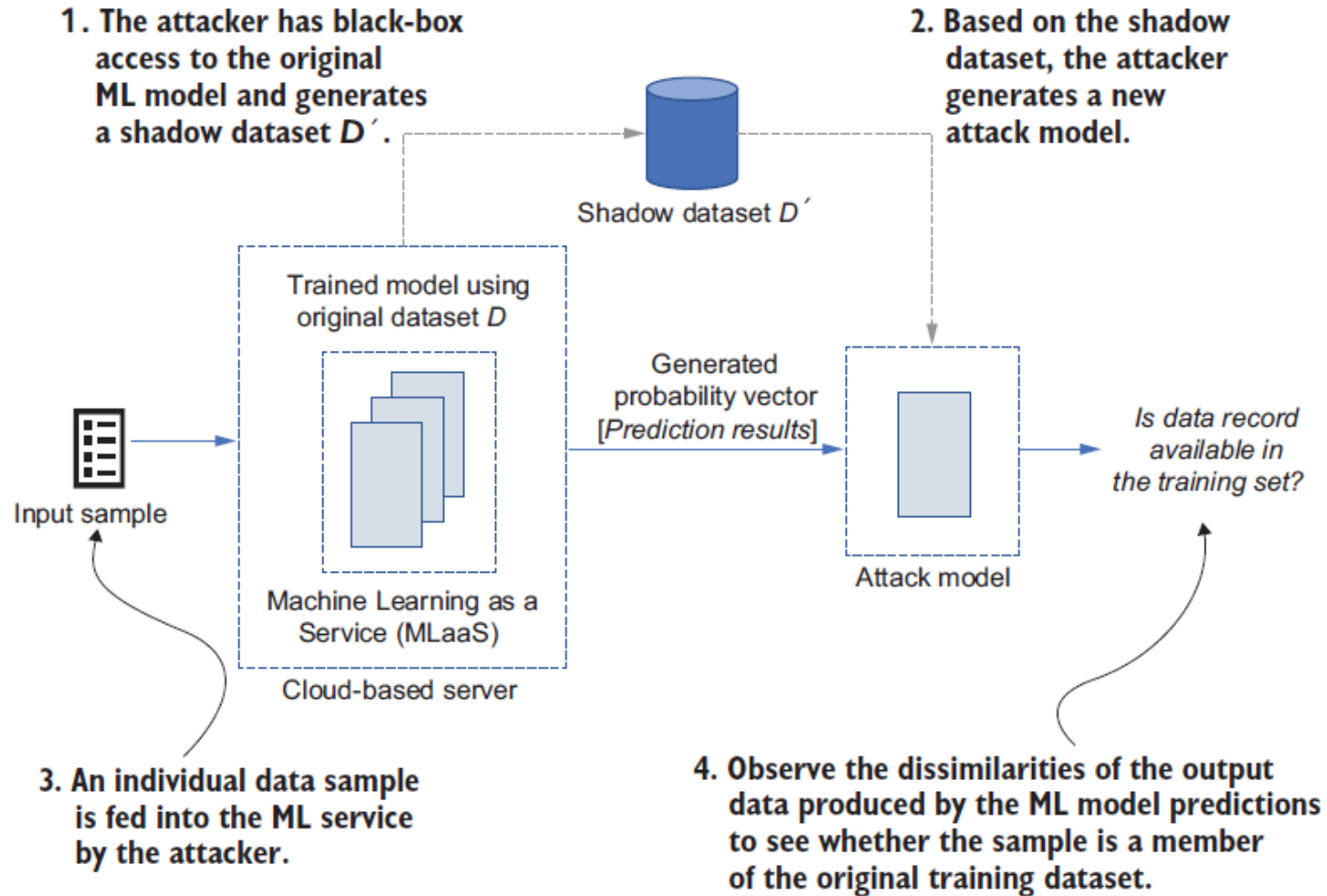


Figure 1.6 How membership inference attack works

Threats and Attacks for ML Systems (Contd...)

Membership inference attacks

- Let's consider an ML-based disease diagnosis system by analyzing the input medical information and conditions.
- For instance, suppose a patient participates in a study that diagnoses the correct difficulty level for a complex game designed to identify people who have Alzheimer's disease.
- If an attacker succeeds in carrying out membership inference, they will know whether this patient was in the original dataset used to build the model.
- Not only that, by knowing the difficulty level of the game, the adversary can deduce whether this patient suffers from Alzheimer's disease.
- This scenario describes is a serious leakage of sensitive information that could be used later for targeted action against the patient.

Threats and Attacks for ML Systems (Contd...)

De-anonymization or re-identification attacks

- Anonymizing datasets before releasing them to third-party users is a typical approach to protecting user privacy.
- In simple terms, anonymization protects private or sensitive information by erasing or changing identifiers that connect an individual to stored data.
- For example, you can anonymize personally identifiable information such as names, addresses, and social security numbers through a data anonymization process that retains the data but keeps the source anonymous.
- Some organizations employ various strategies to release only anonymized versions of their datasets for the general public's use (e.g., public voter databases, Netflix prize dataset, AOL search data, etc.).

Threats and Attacks for ML Systems (Contd...)

- For example, Netflix published a large dataset of 500,000 Netflix subscribers with anonymized movie ratings by inviting different contestants to perform data mining and propose new algorithms to build better movie recommender systems.
- However, even when data is cleared of identifiers, attacks are still possible through de-anonymization.
- De-anonymization techniques can easily cross-reference publicly available sources and reveal the original information.

Threats and Attacks for ML Systems (Contd...)

- In 2008, Narayanan et al. demonstrated that even with data anonymization techniques such as k -anonymity, it is possible to infer the private information of individuals.
- In their attack scenario, they utilized the Internet Movie Database (IMDb) as background knowledge to identify the Netflix records of known users, apparently uncovering users' political preferences.
- Thus, simple syntax-based anonymization cannot reliably protect data privacy against strong adversaries.
- We need to rely on something like differential privacy.

Threats and Attacks for ML Systems (Contd...)

Challenges of privacy protection in big data analytics

- Apart from the threats and attacks tailored explicitly to ML models and frameworks, another privacy challenge arises at the opposite end of the ML and privacy spectrum.
- That is, how can we protect *data at rest*, such as data stored in a database system before it is fed to an ML task, and *data in transit*, where data flows through various elements in the underlying ML framework?
- The ongoing move toward larger and connected data reservoirs makes it more challenging for database systems and data analytics tools to protect data against privacy threats.

Threats and Attacks for ML Systems (Contd...)

- One of the significant privacy threats in database systems is linking different database instances to explore an individual's unique fingerprint.
- This type of attack can be categorized as a subclass of a re-identification attack and is most often an insider attack in terms of database applications.
- Based on the formation and identification of data, these attacks can be further classified into two types:
 - correlation attacks and
 - identification attacks

Threats and Attacks for ML Systems (Contd...)

CORRELATION ATTACKS

- The ultimate purpose of a correlation attack is to find a correlation between two or more data fields in a database or set of database instances to create unique and informative data tuples.
- As you know, in some cases we can bring domain knowledge from external data sources into the identification process.
- For example, let's take a medical records database that lists user information with a history of medication prescriptions.
- Consider another data source with user information along with pharmacies visited.
- Once these sources are linked together, the correlated database can include some additional knowledge, such as which patient bought their medication from which pharmacy.
- Moreover, if it is smart enough to explore the frequently visited pharmacies, an adversary might obtain a rough estimate of where the patient resides.
- Thus, the final correlated dataset can have more private information per user than the original datasets.

Threats and Attacks for ML Systems (Contd...)

IDENTIFICATION ATTACKS

- While a correlation attack tries to extract more private information, an identification attack tries to identify a targeted individual by linking entries in a database instance.
- The idea is to explore more personal information about a particular individual for identification.
- We can consider this one of the most threatening types of data privacy attacks on a dataset as it affects an individual's privacy more.
- For example, suppose an employer looked into all the occurrences of its employees in a medical record or pharmacy customer database.
- That might reveal lots of additional information about the employees' medication records, medical treatments, and illnesses.
- Thus, this attack is an increasing threat to individual privacy.

Threats and Attacks for ML Systems (Contd...)

- At this point, it should be clear that we need to have **sophisticated mechanisms in data analytics and ML applications to protect individuals' privacy** from different targeted attacks.
- Using multiple layers of data anonymization and data pseudonymization techniques makes it possible to protect privacy in such a way that linking different datasets is still possible, but identifying an individual by analyzing the data records is challenging.

Privacy-preserving Machine Learning

Securing privacy while learning from data: Privacy-preserving machine learning

- Many privacy-enhancing techniques concentrate on allowing multiple input parties to collaboratively train ML models without releasing the private data in its original form.
- This collaboration can be performed by utilizing cryptographic approaches (e.g., secure multiparty computation) or differential private data release (perturbation techniques).
- Differential privacy is especially effective in preventing membership inference attacks.
- Finally, the success of model inversion and membership inference attacks can be decreased by limiting the model's prediction output (e.g., class labels only).
- Some of the privacy-enhancing techniques such as differential privacy, local differential privacy (LDP), privacy-preserving synthetic data generation, privacy-preserving data mining techniques, and compressive privacy helps to preserve data privacy.

Privacy-preserving Machine Learning (Contd...)

Use of differential privacy

- The data explosion has resulted in greatly increased amounts of data being held by individuals and entities, such as personal images, financial records, census data, and so on.
- However, the privacy concern is raised when this data leaves the hands of data owners and is used in some computations.
- The AOL search engine log attack and Netflix prize contest attacks demonstrate the existence of such threats and emphasize the importance of having privacy-aware ML algorithm.

Privacy-preserving Machine Learning (Contd...)

- Differential privacy (DP) is a promising solution for providing privacy protection for data.
- It attempts to protect an individual's sensitive information from any inference attacks targeting the statistics or aggregated data of the individual.
- Publishing only the statistics or aggregated data of multiple people in a dataset does not necessarily ensure privacy protection in many cases.
- Let's consider a simple retail use case with a loyalty card scenario.
- Suppose we have two aggregate statistics: the total amount spent by all customers on a particular day and a subgroup of that—the total amount spent by customers using a loyalty card on the same day.
- Suppose there is precisely one customer who purchases without a loyalty card.
- In that case, by simply comparing the difference between two such statistics, someone could easily infer this customer's total amount spent, based on only those aggregate statistics.

Privacy-preserving Machine Learning (Contd...)

- DP is based on the idea that statistics or aggregated data (including ML models) should not reveal whether an individual appears in the original dataset (the training data for the ML models).
- For example, given two identical datasets, one including an individual's information and the other without their information, DP ensures that the probability of generating specific statistics or aggregated values is nearly the same whether conducted on the first or the second dataset.

Privacy-preserving Machine Learning (Contd...)

- To be more specific, consider a trusted data curator that gathers data from multiple data owners and performs a computation on the data, such as calculating the mean value or finding the maximum or minimum value.
- To ensure no one can reliably infer any individual sample from the computation result, DP requires the curator to add random noise to the result, such that the released data will not change if any sample of the underlying data changes.
- Since no single sample can significantly affect the distribution, adversaries cannot confidently infer the information corresponding to any individual sample.
- Thus, a mechanism is differentially private if the computation result of the data is robust to any change in the individual samples.
- Due to its underlying mechanisms, DP techniques resist membership inference attacks by adding random noise to the input data, to iterations of a particular ML algorithm, or to algorithm output.

Privacy-preserving Machine Learning (Contd...)

Local differential privacy

- When the input parties do not have enough information to train an ML model, it can be better to utilize approaches that rely on local differential privacy (LDP).
- For instance, multiple cancer research institutions want to build an ML model to diagnose skin lesions, but no single party has enough data to train a model.
- LDP is one of the solutions that they can use to train an ML model collaboratively without violating individual privacy.
- In LDP, individuals send their data to the data aggregator after privatizing data by perturbation.
- Hence, these techniques provide plausible deniability (an adversary cannot prove that the original data exists) for individuals.

Privacy-preserving Machine Learning (Contd...)

- The data aggregator collects all the perturbed values and estimates statistics such as the frequency of each value in the population.
- Compared with DP, LDP shifts the perturbation from the central site to the local data owner.
- It considers a scenario where there is no trusted third party and an untrustworthy data curator needs to collect data from data owners and perform certain computations.
- The data owners are still willing to contribute their data, but the privacy of the data must be enforced.
- An old and well-known version of local privacy is a randomized response (RR), which provides plausible deniability for respondents to sensitive queries.

Privacy-preserving Machine Learning (Contd...)

- For example, a respondent flips a fair coin:
 - If tails, the respondent answers truthfully
 - If heads, they flip a second coin and respond “yes” if heads and “no” if tails
- An ML-oriented work, AnonML, utilized the ideas of RR to generate histograms from multiple input parties.
- AnonML uses these histograms to create synthetic data on which an ML model can be trained. Like other LDP approaches, AnonML is a good option when no input party has enough data to build an ML model on their own (and when there is no trusted aggregator).

Privacy-preserving Machine Learning (Contd...)

Privacy-preserving synthetic data generation

- Although many privacy-preserving techniques have been proposed and developed for all kinds of ML algorithms, sometimes data users may want to execute new ML algorithms and analysis procedures.
- When there is no predefined algorithm for the requested operation, data users may request data to utilize it locally.
- To that end, different privacy-preserving data-sharing techniques such as k -anonymity, l -diversity, t -closeness, and data perturbation have been proposed in the past.
- These techniques can be fine-tuned to generate a new anonymized dataset from the same original dataset.
- However, in some cases, anonymization alone may hurt the utility of the underlying ML algorithms. Thus, a promising solution for data sharing is to generate synthetic yet representative data that can be safely shared with others.

Privacy-preserving Machine Learning (Contd...)

- Synthetic data is generated artificially rather than produced through real-world events at a high level.
- It is usually generated algorithmically and is often used as a stand-in for training and testing ML models.
- Nevertheless, in practice, sharing a synthetic dataset in the same format (preserving the same statistical characteristics) as the original dataset gives data users much more flexibility with minimal privacy concerns.

Privacy-preserving Machine Learning (Contd...)

Privacy-preserving data mining techniques

- Among privacy-preserving data mining (PPDM) techniques, the vast majority rely on either modifying data or removing some of the original content to protect privacy.
- The resulting quality degradation from this sanitization or transformation is the tradeoff between the quality of data and the level of privacy.
- Nevertheless, the basic idea behind all these PPDM techniques is to efficiently mine data while preserving the privacy of individuals.
- There are three main classes of techniques for dealing with PPDM, based on the different stages of data collection, publishing, and processing.

Privacy-preserving Machine Learning (Contd...)

TECHNIQUES FOR PRIVACY-PRESERVING DATA COLLECTION

- The first class of PPDM techniques ensures privacy at the stage of data collection.
- It usually incorporates different randomization techniques at the data collection stage and generates privatized values, so original values are never stored.
- The most common randomization approach is to modify data by adding some noise with a known distribution.
- Whenever data mining algorithms are involved, the original data distribution, but not the individual values, can be reproduced.
- Additive and multiplicative noise approaches are two of the most common data randomization techniques in this category.

Privacy-preserving Machine Learning (Contd...)

DIFFERENT APPROACHES TO DATA PUBLISHING AND PROCESSING

- The second class of PPDM deals with techniques related to when data is released to third parties (published) without disclosing the ownership of sensitive information.
- Removing attributes that can explicitly identify an individual from a dataset is not sufficient, as users may still be identified by combining nonsensitive attributes or records.

Privacy Preserving Techniques

- Many privacy-enhancing techniques concentrate on allowing multiple input parties to collaboratively train ML models without releasing the private data in its original form.
- This collaboration can be performed by utilizing cryptographic approaches (e.g., secure multiparty computation) or differential private data release (perturbation techniques).
- Differential privacy is especially effective in preventing membership inference attacks.

Differential Privacy

- Popular privacy protection scheme used in applications today
- A mathematical technique of adding a controlled amount of randomness to dataset
- Prevents anyone from obtaining information about individuals
- Maintains accuracy for aggregate insights
- Balances data utility and privacy

Differential Privacy (Contd...)

- The name "differential privacy" reflects the core idea behind this privacy framework: ensuring that the presence or absence of any single individual's data makes little difference in the overall output of a query.
- The term emphasizes the "differential" aspect—how much the output changes when a single record is added or removed.
- "Differential" - The term originates from the mathematical concept of sensitivity (or difference) in outputs when one data point changes.
- It measures the extent to which an algorithm's output can change when a single individual's data is modified.
- "Privacy" - The goal is to ensure that an adversary cannot infer sensitive information about any individual in the dataset, even if they have external knowledge.

Differential Privacy (Contd...)

Why is it called Differential Privacy?

- It guarantees that for two nearly identical datasets (differing by only one entry), the probability distribution of query results remains statistically similar.
- This means an adversary cannot confidently determine whether a particular individual's data was included in the dataset.
- The privacy guarantee is formalized using the ϵ -differential privacy condition:

$$\Pr[M(D)] \leq e^\epsilon \Pr[M(D')]$$

"The probability that mechanism M produces a given output on dataset D is at most e^ϵ times the probability that it produces the same output on a neighboring dataset D'

where: $M(D)$ is the mechanism (query function) applied to dataset D , D' is a dataset differing from D by at most one record, e is just the mathematical constant 2.718, ϵ is the privacy budget (smaller values mean stronger privacy).

Differential Privacy (Contd...)

The Laplace distribution plays a crucial role in differential privacy by enabling the addition of controlled noise to data queries, ensuring privacy while maintaining useful statistical properties.

Connection Between Laplace Distribution and Differential Privacy

1. Noise Addition for Privacy

- Differential privacy ensures that the presence or absence of a single record in a dataset does not significantly affect the output of a query.
- To achieve this, noise is added to the query result, making it difficult to infer individual data points.
- The Laplace distribution is commonly used to generate this noise.

2. Laplace Mechanism

- The Laplace Mechanism is a technique in differential privacy where noise is sampled from a Laplace distribution and added to query results.

Differential Privacy – Laplace Distribution

The **Laplace distribution** is like a sharper, more "pointy" version of the normal (bell curve) distribution.

It looks like a peak in the middle and has "fatter" tails compared to the normal distribution.

This means that most values are clustered around the center, but extreme values (outliers) are more common than in a normal distribution.

The Laplace distribution is defined as:

$$f(x|b)=1/2b.\exp(-|x|/b)$$

"f of x given b is equal to one over two b, times e to the power of negative absolute x over b."

where μ (mu) is the middle (peak) of the distribution, also called the "location" parameter and b is the scale parameter.

Differential Privacy

The Laplace distribution is defined as:

$$f(x|b)=1/2b.\exp(-|x|/b)$$

where b is the scale parameter.

The scale b is set based on the sensitivity of the query function and the desired privacy level ϵ :

$$b=\Delta f/\epsilon$$

where: Δf is the sensitivity of the function (maximum change in output due to a single data point),

ϵ is the privacy budget, controlling the level of privacy.

Differential privacy: When adding noise, Laplace ensures rare, larger jumps in noise, making it harder to detect original data.

Differential Privacy

Why Laplace Distribution?

- The Laplace distribution is centered at zero and has exponentially decaying tails, ensuring that small noise values are more likely while still allowing for occasional larger noise values.
- It satisfies ϵ -differential privacy, meaning that the probability of obtaining a certain output with or without a particular data point in the dataset differs only by a factor of e^ϵ .
- The Laplace distribution is fundamental to achieving differential privacy.
- The Laplace Mechanism ensures that queries remain differentially private by injecting noise based on sensitivity and privacy parameters.
- The scale parameter $b = \Delta f / \epsilon$ controls the trade-off between privacy and accuracy.

Local Differential Privacy

- When the input parties do not have enough information to train an ML model, it can be better to utilize approaches that rely on local differential privacy (LDP).
- For instance, multiple cancer research institutions want to build an ML model to diagnose skin lesions, but no single party has enough data to train a model.
- LDP is one of the solutions that they can use to train an ML model collaboratively without violating individual privacy.
- In LDP, individuals send their data to the data aggregator after privatizing data by perturbation.
- Hence, these techniques provide plausible deniability (an adversary cannot prove that the original data exists) for individuals.

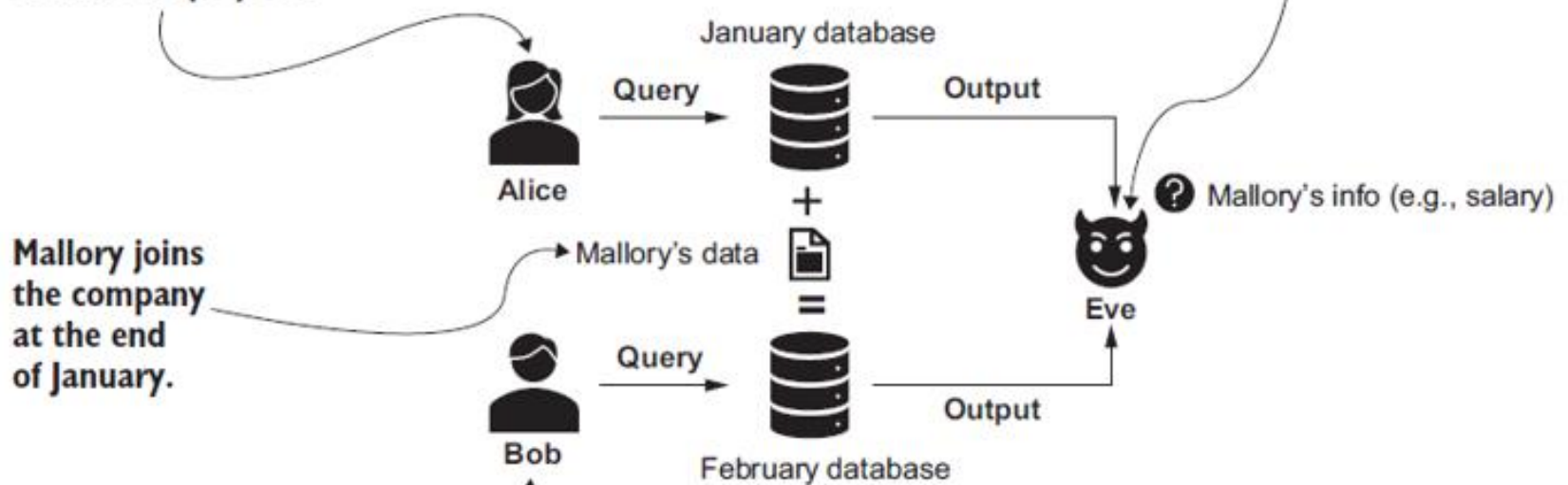
Local Differential Privacy

- The data aggregator collects all the perturbed values and estimates statistics such as the frequency of each value in the population.
- Compared with DP, LDP shifts the perturbation from the central site to the local data owner.
- It considers a scenario where there is no trusted third party and an untrustworthy data curator needs to collect data from data owners and perform certain computations.
- The data owners are still willing to contribute their data, but the privacy of the data must be enforced.
- An old and well-known version of local privacy is a randomized response (RR), which provides plausible deniability for respondents to sensitive queries.

Personal Information Leakage

In January, news reporter Alice queries the average salary of a private company from its database, which contains personal information (e.g., salaries) of all its employees.

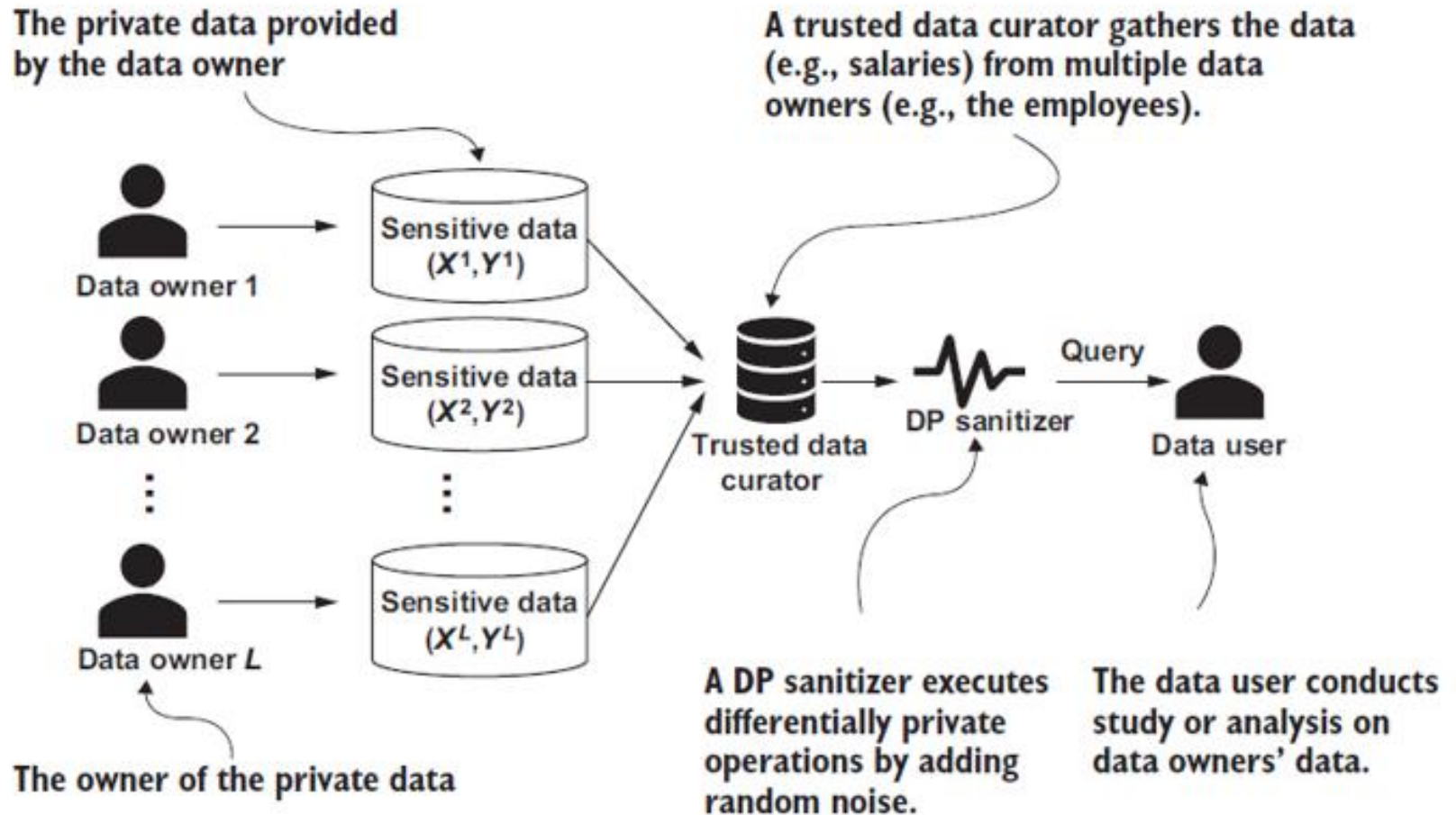
News reporter Eve learns Mallory's info (e.g., salary) by comparing Alice's report (before Mallory joins) and Bob's report (after Mallory joins).



Mallory joins the company at the end of January.

In February, news reporter Bob queries the average salary of a private company from its database, which contains personal information (e.g., salaries) of all its employees.

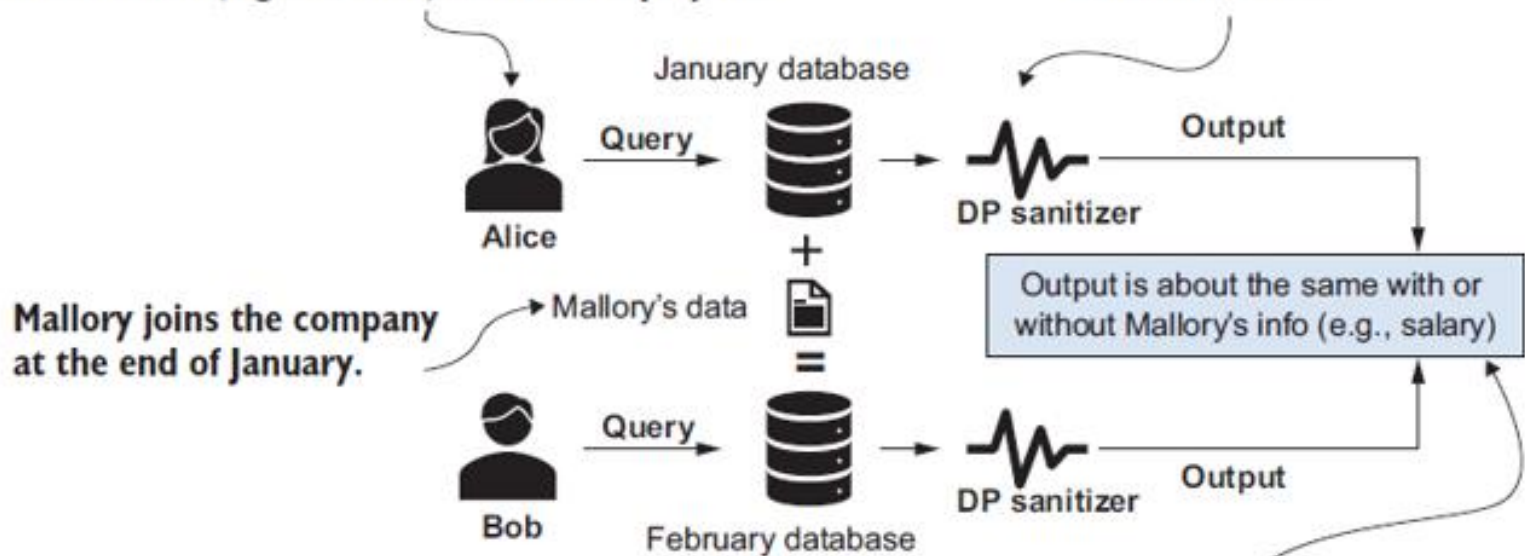
Differential Privacy Framework



DP to protect personal data

In January, news reporter Alice queries the average salary of a private company from its database, which contains personal information (e.g., salaries) of all its employees.

DP sanitizer executes differentially private operations by adding random noise.



In February, news reporter Bob queries the average salary of a private company from its database, which contains personal information (e.g., salaries) of all its employees.

DP ensures that the probability of getting the same aggregated or statistical information or of reaching a specific conclusion would be the same, no matter which database the analysis is conducted on.

Differential Privacy

Sensitivity and Privacy Budget

- How much noise to be added is based on **sensitivity** and **privacy budget**
- Challenge is to determine the amount of random noise to be added to the aggregated data
- Noise cannot come from an arbitrary random variable
- Noise is too small - cannot provide enough protection
- Noise is too large - published aggregated data would be distorted and meaningless; provides no utility
- DP requires that the output of the analysis remain “approximately the same” but not “exactly the same” (e.g., both with and without Mallory’s information)

Differential Privacy (Contd...)

Sensitivity in DP

- Sensitivity - measures the maximum impact a single individual's data can have on the output of a query or function.
- Purpose - determines how much noise needs to be added to ensure privacy.
- Calculation - Sensitivity is the difference between the results of a query on two datasets differing by only one individual.

Privacy Budget in DP

- Privacy budget (often denoted by epsilon, ϵ) quantifies the level of privacy protection.
- It controls the trade-off between privacy and data utility (accuracy).
- Smaller Budget (Low ϵ): Provides stronger privacy but reduces the accuracy of results.
- Larger Budget (High ϵ): Allows more accurate results but weakens privacy

Differential Privacy (Contd...)

Example Scenario: Average Salary Calculation

1. **Dataset:** A company's dataset contains the salaries of employees, e.g., [50,000, 60,000, 70,000, 80,000].
2. **Goal:** Compute the average salary while ensuring privacy, so no individual's salary can be inferred.
3. **Query:** The **average salary** is
Average=Sum of Salaries/Number of Employees

The **true average salary** is:

$$\text{True Average} = (50,000 + 60,000 + 70,000 + 80,000) / 4 = 65,000$$

Differential Privacy (Contd...)

Step 1:

Calculating Sensitivity for Average Salary

When one employee is **removed** from the dataset, the sum and the number of employees change, affecting the average. Sensitivity (Δf) measures the maximum possible change in the average.

1. Remove the employee with the **lowest salary (50,000)**:

Remaining dataset:

[60,000,70,000,80,000]

New Average = $(60,000 + 70,000 + 80,000) / 3 = 70,000$

Change in average:

$|70,000 - 65,000| = 5,000$

2. Remove the employee with the **highest salary (80,000)**:

Remaining dataset: [50,000,60,000,70,000]

New Average = $(50,000 + 60,000 + 70,000) / 3 = 60,000$

Change in average: $|65,000 - 60,000| = 5,000$

Sensitivity (Δf):

The largest possible change in the average occurs in this scenario:

$\Delta f = \text{Maximum Salary} - \text{Minimum Salary} / \text{Number of Employees}$

For this example:

$\Delta f = 80,000 - 50,000 / 4 = 30,000 / 4 = 7,500$

Differential Privacy (Contd...)

Step 2: Add Noise Based on ϵ

From Step 1, we calculated

$$\Delta f = \text{Maximum Salary} - \text{Minimum Salary} / n = 7,500$$

Now, we use this sensitivity (Δf) to add noise. The scale of the noise is determined by:

$$b = \Delta f / \epsilon$$

Δf : Sensitivity of the query.

ϵ : Privacy budget (a small positive value that controls the privacy-accuracy tradeoff).

Differential Privacy (Contd...)

Adding Noise for Different ϵ Values

The true average salary is 65,000, and we need to add noise based on ϵ .

1. Small $\epsilon=0.1$ (High Privacy):

$$b = \Delta f / \epsilon = 7,500 / 0.1 = 75,000$$

Noise is drawn from $\text{Laplace}(0, 75,000)$. This means noise values might vary significantly (e.g., $\pm 50,000$ or more).

Result:

Noisy average = 65,000 + Noise (e.g., $65,000 + 50,000 = 115,000$ or $65,000 - 50,000 = 15,000$).

Effect: Strong privacy, but the noisy result may be far from the true average.

Differential Privacy (Contd...)

2. Moderate $\epsilon=1$ (Balanced Privacy):

$$b = \Delta f / \epsilon = 7,500 / 1 = 7,500.$$

Noise is drawn from $\text{Laplace}(0, 7,500)$. Noise values are moderate (e.g., $\pm 5,000$ or $\pm 7,000$).

Result:

Noisy average = $65,000 + \text{Noise}$ (e.g., $65,000 + 5,000 = 70,000$ or $65,000 - 5,000 = 60,000$).

Effect: Privacy is preserved while the result remains reasonably accurate.

Differential Privacy (Contd...)

3. Large $\epsilon=5$ (Low Privacy):

$$b = \Delta f / \epsilon = 7,500 / 5 = 1,500.$$

Noise is drawn from $\text{Laplace}(0, 1,500)$. Noise values are minimal (e.g., $\pm 1,000$ or $\pm 2,000$).

Result:

Noisy average = $65,000 + \text{Noise}$ (e.g., $65,000 + 1,000 = 66,000$ or $65,000 - 1,000 = 64,000$).

Effect: High accuracy, but privacy guarantees are weaker.

Differential Privacy (Contd...)

Step 3: Interpretation of ϵ The choice of ϵ reflects the desired balance between privacy and accuracy:

Small ϵ : Strong privacy, significant noise, low utility (results might vary widely).

Moderate ϵ : Balanced trade-off, good for most practical applications.

Large ϵ : Weak privacy, minimal noise, high utility.